CONFLUENT AI DAY INDIA HACKATHON 2026

scam-shield

Production-Grade Real-Time Payment Fraud Defence

The bet no bank has been willing to make: put AI where it cannot do regulatory harm.

< 350ms
End-to-End Decision

Dedicated
Private Link Cluster

9 SQL
Flink Statements

3 LLMs
Gemini·GPT-4o·Granite

PRODUCER
HMAC-SHA256

KAFKA
Private Link

FLINK ★
9 SQL

DECIDE
350ms

GOVERNED AI
Narrate

IBM Stack · Confluent Cloud · HashiCorp Terraform · Azure · RBIH MuleHunter.AI · Google Gemini · IBM WatsonX

BUSINESS CASE

₹36,014 Cr

Lost to bank fraud in a single year.

Source: RBI Annual Report FY 2024-25

The problem is accelerating — not slowing. Every year without production-grade defence is another ₹36,000 crore.

Sources: RBI Annual Report FY25 · RBI H1 FY25 · Ministry of Finance Lok Sabha 2025

194%
Surge in fraud value FY24-25

8×
Jump in banking fraud H1 FY25

1.13M
Cyber fraud cases registered 2023

56%
UPI share of all digital fraud

WHY IT STAYS UNSOLVED

Technology isn't the gap. The architecture is.

Speed vs Accuracy

UPI clears in milliseconds. Batch ML scoring is too slow. Banks choose speed — and accept more fraud.

PII + AI = Regulatory Trap

Sending customer data to cloud AI violates RBI data localisation. Banks can't use powerful AI safely.

Black-Box Decisions

ML models cannot explain a BLOCK to an RBI examiner. No explainability means no regulatory defensibility.

Siloed Payment Rails

Fraud jumps UPI → IMPS → Card in seconds. Per-rail tools built in isolation miss coordinated attacks.

No Human-in-the-Loop at Scale

Thousands of daily alerts bury the most dangerous cases. Analysts cannot review what they cannot find.

THE ARCHITECTURE BET

Flink decides.
AI explains.

The governance architecture that makes streaming AI safe to deploy in a regulated bank — not deploying a model, deploying the framework that makes it defensible.

INV-1
No raw PII ever reaches a model

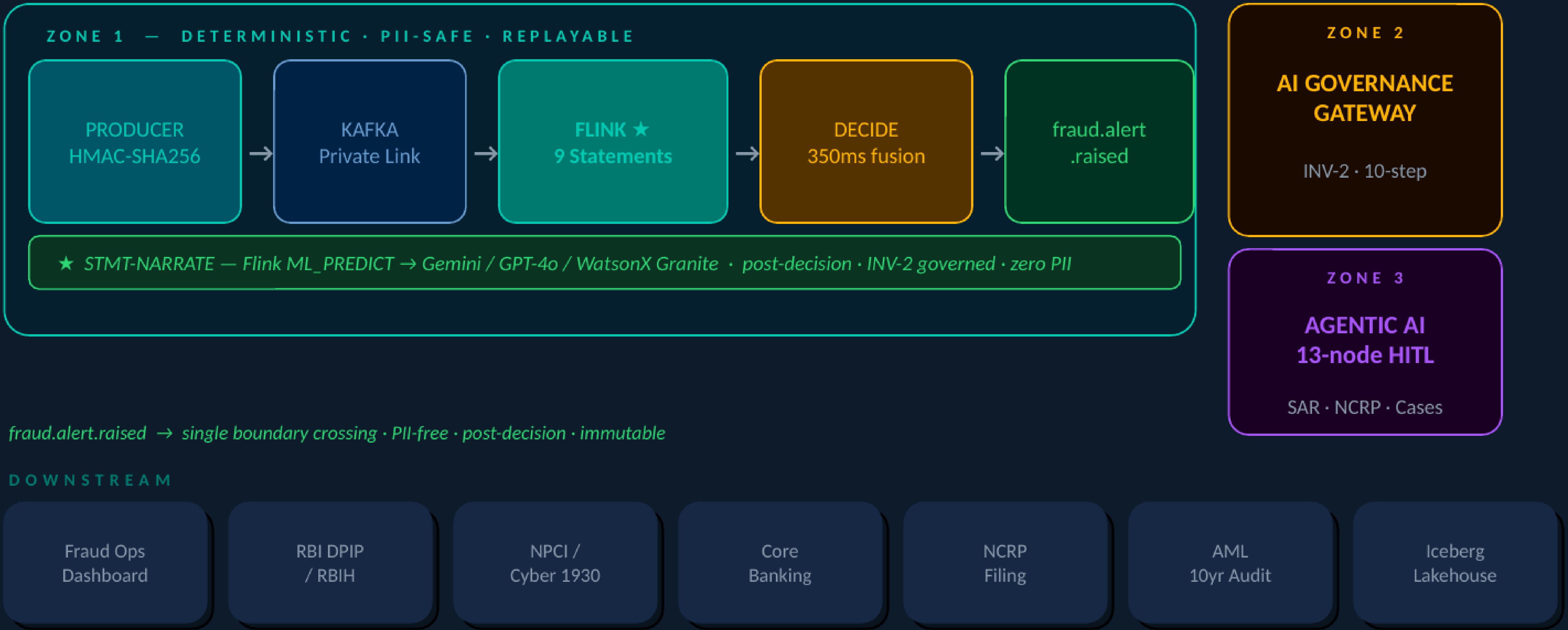
INV-2
Every model call is governed & audited

INV-3
Decision is deterministic & replayable

INV-4
All topics follow domain.entity.event

PLATFORM ARCHITECTURE

Three Zones · One Decision Boundary

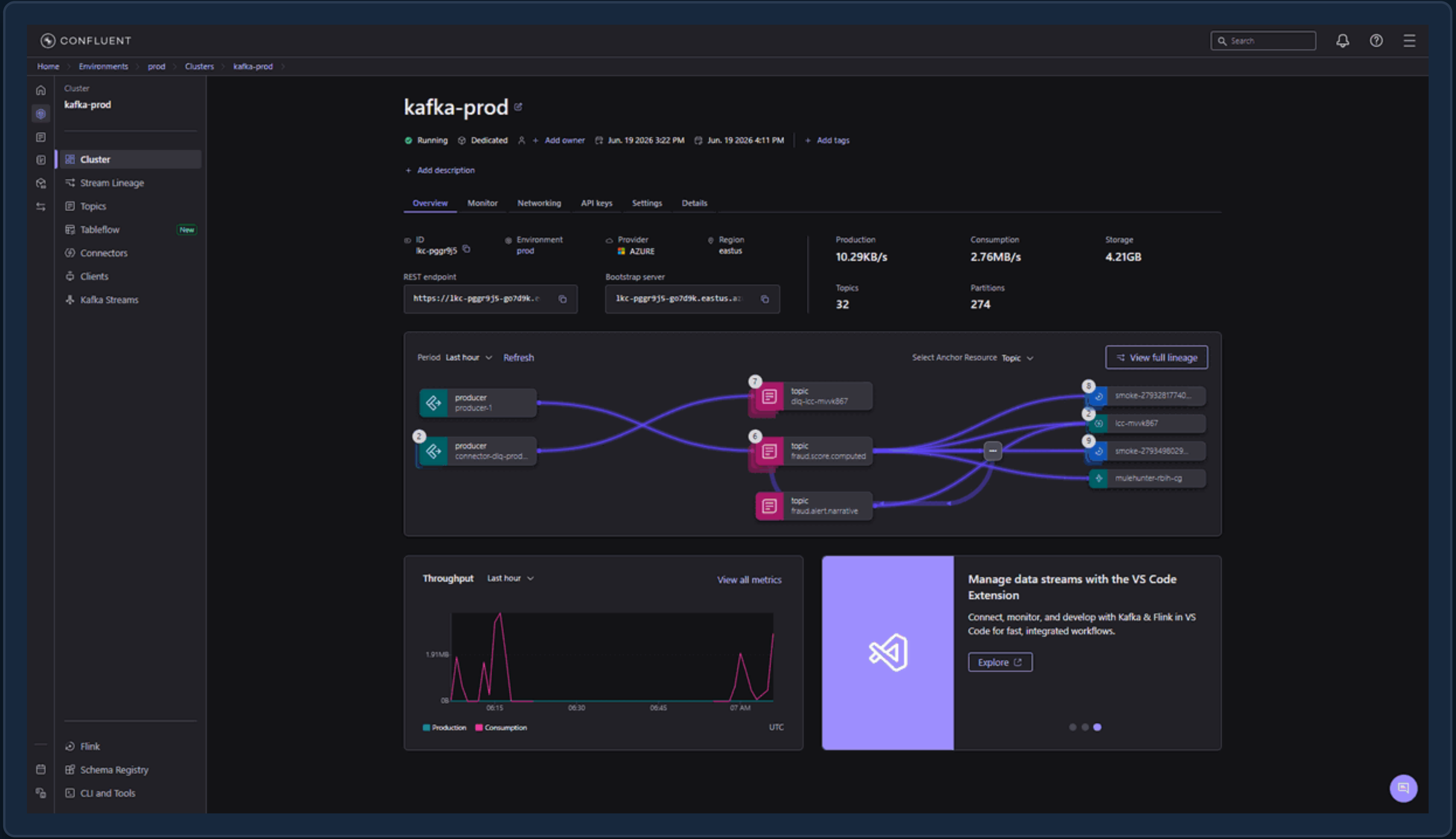


Tier-1 P99 < 120ms (hard-block / Flink-only) · Tier-2 P99 < 350ms (full ensemble)

PLATFORM

A dedicated, private cluster on Azure East US

● LIVE



WHAT YOU'RE SEEING

- Dedicated cluster on Private Link, status Running
- Provider Azure · region eastus · 274 partitions
- 32 topics, ~4.2 GB stored, live throughput shown
- Stream Lineage maps producers → topics → consumers

WHY IT MATTERS *Enterprise-grade isolation with an auditable, replayable event backbone — not a demo cluster.*

CONFLUENT FLINK · THE HERO

9 SQL Statements. 15 Logical Jobs. The Deterministic Detection Core.

< 120ms	< 350ms	9	15	~40%
Tier-1 P99	Tier-2 P99	Statements	Logical Jobs	CFU saved

STMT-INGEST	Rail filter · dedup · sanity check	JOB-01/02/03
STMT-ENRICH	11 temporal joins · velocity · behavioral	JOB-04
STMT-RULES	Per-rail CEP · MATCH_RECOGNIZE	JOB-05/06/07
STMT-CEPXR	Cross-rail CEP · ATO + Mule chain	JOB-08
STMT-DECIDE ★	350ms fusion · 40/40/20 · hard-block first	JOB-09/10/12
STMT-COMPLAINT	30-day interval join · preventive HOLD	JOB-11
STMT-NARRATE ★	ML_PREDICT → Gemini / GPT-4o / WatsonX	AI overlay

SCORE FUSION

ensemble =

Flink Rules × 40%

MuleHunter.AI × 40%

AML Engine × 20%

Decision window: 350ms tumbling

Hard-block fires FIRST — bypasses ensemble

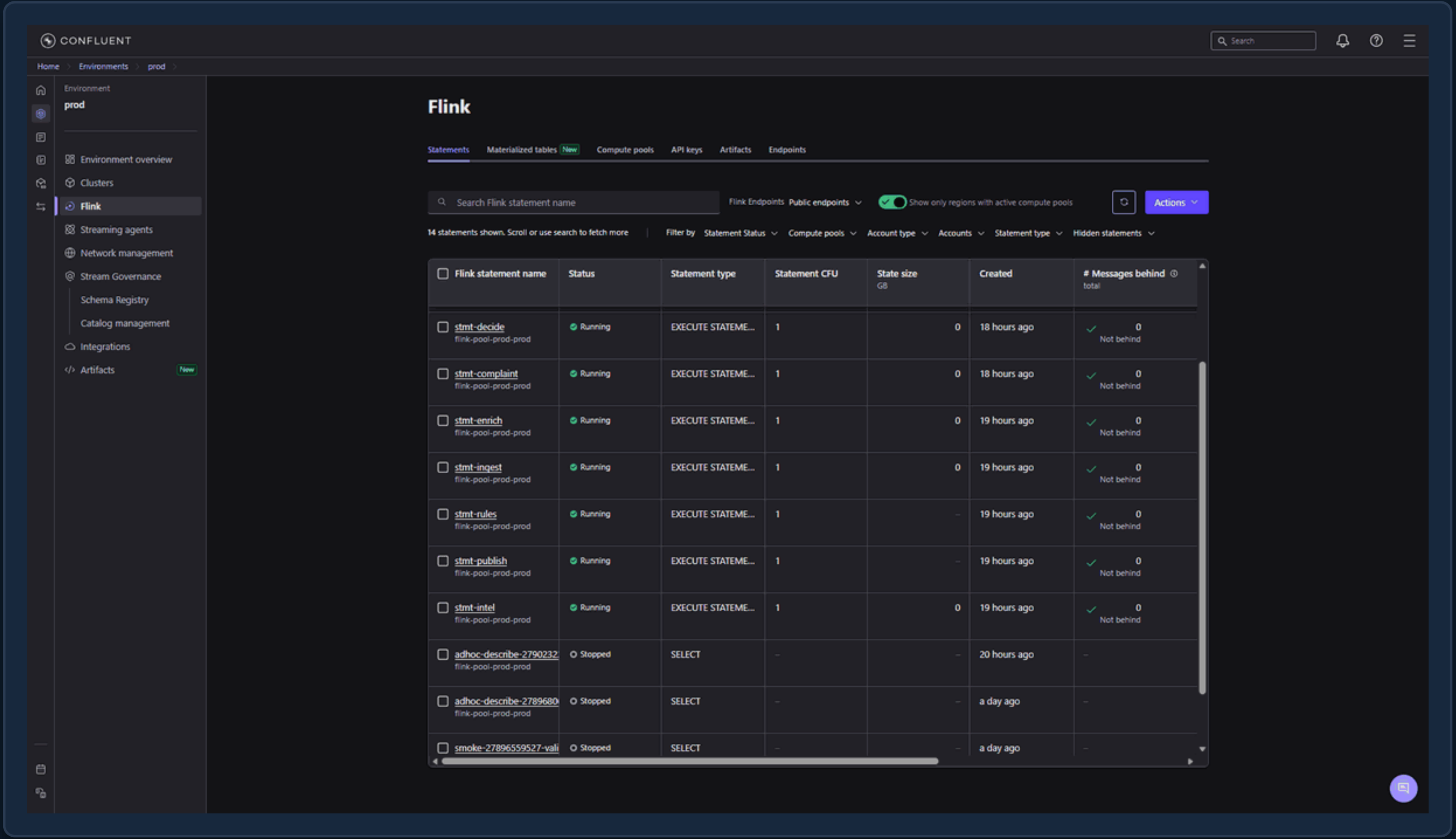
Tier-1 fast path: < 120ms P99

Tier-2 full: < 350ms P99

FLINK CORE

The nine SQL statements, running live

● LIVE



WHAT YOU'RE SEEING

- ingest → enrich → rules → cepxr → decide → narrate
- plus stmt-publish, stmt-intel and stmt-complaint
- all Running · ~1 CFU each · zero messages behind
- one shared, elastically-scaled compute pool

WHY IT MATTERS *The 9 SQL statements from the architecture slide — deployed and observable, not slideware.*

THE CODE THAT MAKES IT REAL

STMT-CEPXR + STMT-DECIDE · Deterministic. Explainable. Replayable.

STMT-CEPXR — XR_ATO_PROBE_AND_DRAIN

```
SELECT * FROM
  `payments.transaction.enriched`
MATCH_RECOGNIZE (
  PARTITION BY account_id_hash
  ORDER BY `$rowtime`
  MEASURES
    LAST(BIG.txn_id)  AS txn_id,
    FIRST(PROBE.`$rowtime`) AS start
  ONE ROW PER MATCH
  AFTER MATCH SKIP TO NEXT ROW
  PATTERN (PROBE BIG)
  WITHIN INTERVAL '5' MINUTE
  DEFINE
    PROBE AS PROBE.amount < 100,
    BIG   AS BIG.amount > 10000
       AND BIG.bene_txn_count_30d = 0
);

-- XR_MULE_CHAIN: 3+ credits → debit
  PATTERN (CRD{3,} OUTB)
  WITHIN INTERVAL '10' MINUTE
  DEFINE CRD AS direction='CREDIT',
    OUTB AS direction='DEBIT'
      AND bene_txn_count_30d = 0
```

STMT-DECIDE — Score Fusion (350ms)

```
-- 40/40/20 weighted ensemble
ensemble_score =
  0.40 * flink_rules_score
+ 0.40 * mulehunter_score
+ 0.20 * COALESCE(aml_score,
    behavioral_score, 0.0)

-- 350ms tumbling window
FROM TABLE(TUMBLE(
  TABLE `fraud.score.computed`,
  INTERVAL '0.350' SECOND))

-- Thresholds (priority order)
HARD_BLOCK override → first
rules_score >= 0.80  → BLOCK
ensemble    >= 0.70  → BLOCK
ensemble    >= 0.60  → CHALLENGE
ensemble    >= 0.40  → HOLD
else                 → ALLOW

-- 6 atomic sinks (STATEMENT SET)
fraud.alert.raised
+ fraud.account.freeze
+ platform.ops.event
```

STMT-NARRATE · FLINK ML_PREDICT

Three LLM Backends.
One Governed Topic. Zero PII. Post-Decision Only.

INV-2 · every call governed

INV-3 · runs after DECIDE

Zero PII · post-decision

Google Gemini

PRIMARY

googleai provider

Native Confluent Flink provider. Zero proxy. Publicly reachable from Confluent Cloud.

```
CREATE MODEL `gemini_narrate`
WITH (
  'provider'          = 'googleai',
  'googleai.connection' = 'gemini-conn',
  'googleai.system_prompt' =
    'You are a fraud analyst...'
);

-- confluent flink connection create
  gemini-conn --type googleai
```

Azure OpenAI GPT-4o

SUPPORTED

azureopenai provider

Native Confluent Flink azureopenai provider. Azure auth handled natively — zero proxy.

```
CREATE MODEL `azure_openai_narrate`
WITH (
  'provider'           = 'azureopenai',
  'azureopenai.connection'=
     'azure-openai-conn',
  'azureopenai.system_prompt' =
    'You are a fraud analyst...'
);

-- confluent flink connection create
  azure-openai-conn --type azureopenai
```

IBM WatsonX Granite

SUPPORTED

openai-compat proxy

OpenAI-compat proxy on Azure Container Apps. Auto IBM IAM token refresh + project_id injection.

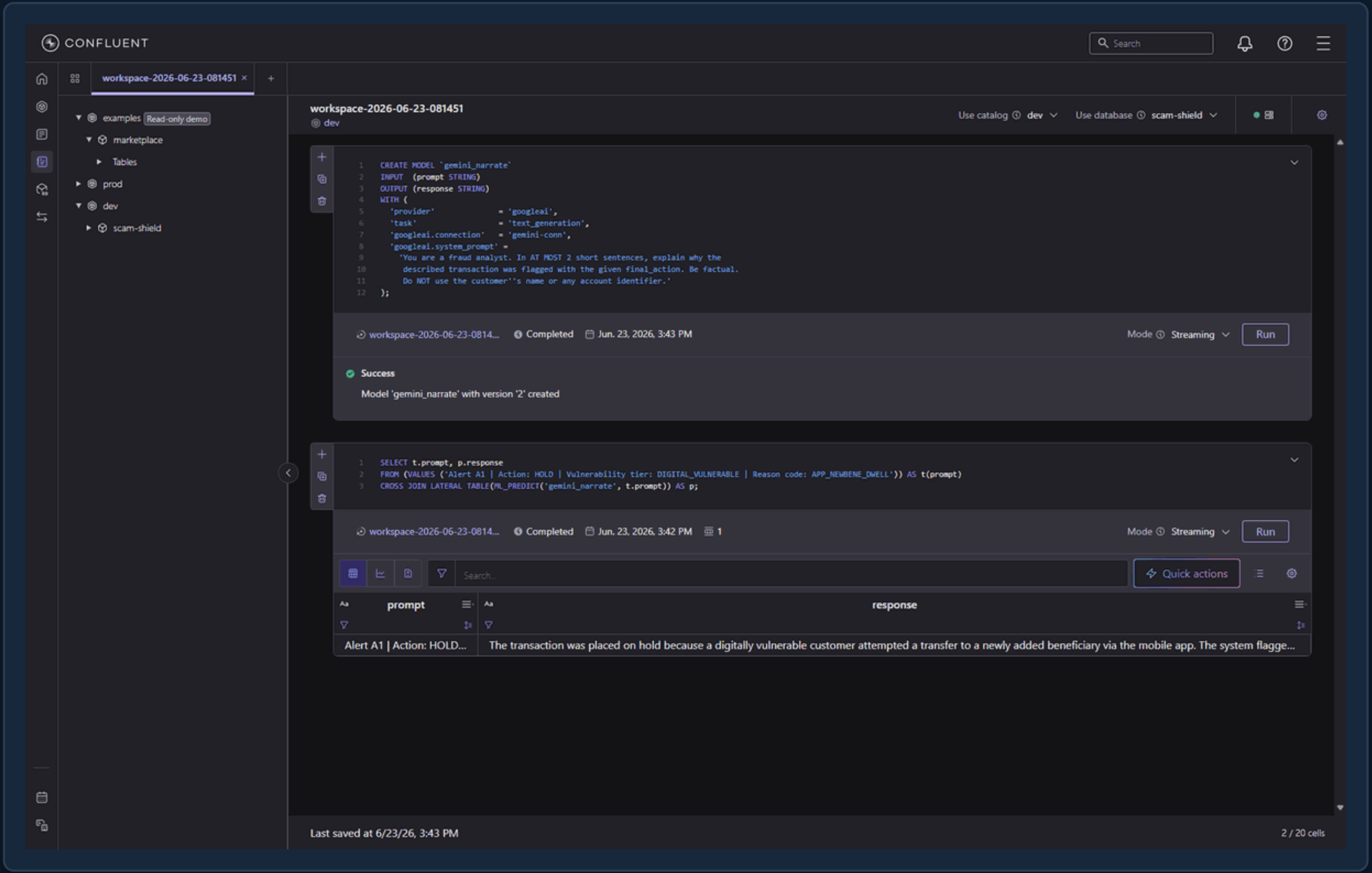
```
CREATE MODEL `watsonx_narrate`
WITH (
  'provider'          = 'openai',
  'openai.connection' = 'watsonx-conn',
  'openai.model_version' =
     'ibm/granite-4-h-small'
);

-- Proxy on Azure Container Apps
-- IBM IAM auto-refresh (~55 min)
-- project_id injection
```

```
All 3 → ML_PREDICT('model', CONCAT(alert_id, ' | Action: ', final_action, ' | Vuln: ', vulnerability_tier)) →
fraud.alert.narrative  ·  model_id stamps provenance
```

GOVERNED AI

LIVE

AI narration runs inside Flink — live



WHAT YOU'RE SEEING

- CREATE MODEL gemini_narrate (Google Gemini backend)
- ML_PREDICT invoked straight from Flink SQL
- A HOLD alert → a plain-English analyst note in seconds
- Empathetic, PII-free and stamped with the model id

WHY IT MATTERS *STMT-NARRATE in action — post-decision AI explains the alert; it never makes the decision (INV-3).*

AGENTIC AI · AI GOVERNANCE

AI that Assists. Never Decides. Always Audited.

LANGGRAPH — 13 NODES

orchestrator_node	validate + route
context_enrichment_node	Qdrant RAG + GPT-4o
upi / card / transfer nodes	specialist scorers ×3
pattern_gap_node ★	Learning Loop F3
decision_node ⚡ HITL	interrupt_before gate
case_management_node	SAR draft + queue
notification_node	PMLA §8A templates
compliance_node	NCRP + RBI DPIP
sar_generation_node	Multi-signal SAR draft
knowledge_update_node ★	Learning Loop F1
output_node	ops.event + gov audit

HITL POLICY

HARD BLOCK
Auto-freeze (any amount) · flag_ncrp always HITL

BLOCK ≤₹5L ≥0.85
Auto-freeze + notify · flag_ncrp HITL

BLOCK > ₹5L
Notify only · freeze always HITL

SAR required
ALL actions → HITL · No exceptions

CHALLENGE ≥0.85
Auto send_otp_challenge only

HOLD any
All actions → HITL

4 MCP SERVERS

- fraudscoringmcpserver :8005
- fraudinvestigationmcpserver :8009
- amltriagemcpserver :8010
- fraudactionmcpserver :5002

CONNECTORS · TABLEFLOW

The Data Contracts That Make It Production-Ready

MANAGED CONNECTORS

PostgresCdcSourceV2
SOURCE

10 compacted lookup topics (account baseline, sanctions, velocity…) · Debezium pgoutput · TopicRegexRouter SMT · feeds FOR SYSTEM_TIME AS OF in STMT-ENRICH

PostgresSink — Alerts
SINK

fraud.alert.raised + fraud.alert.narrative → sinks.fraud_alerts + fraud_alert_narratives · UPSERT on alert_id · Dashboard AI Analyst Notes join

PostgresSink — Ops
SINK

platform.ops.event → sinks.ops_events · ReplaceField$Value SMT drops nested Avro STRUCT fields before insert

CONFLUENT TABLEFLOW → APACHE ICEBERG

Zero ETL · 4 analytical topics materialised to open Iceberg lakehouse
1 declarative config replaces 4 sink connectors

fraud.features.inference
→ ML Feature Store / AI Inference

payments.graph.edges
→ Graph DB staging + analytics

fraud.intelligence.dpip
→ NPCI DPIP regulatory export

fraud.intelligence.mulehunter
→ MuleHunter retraining corpus

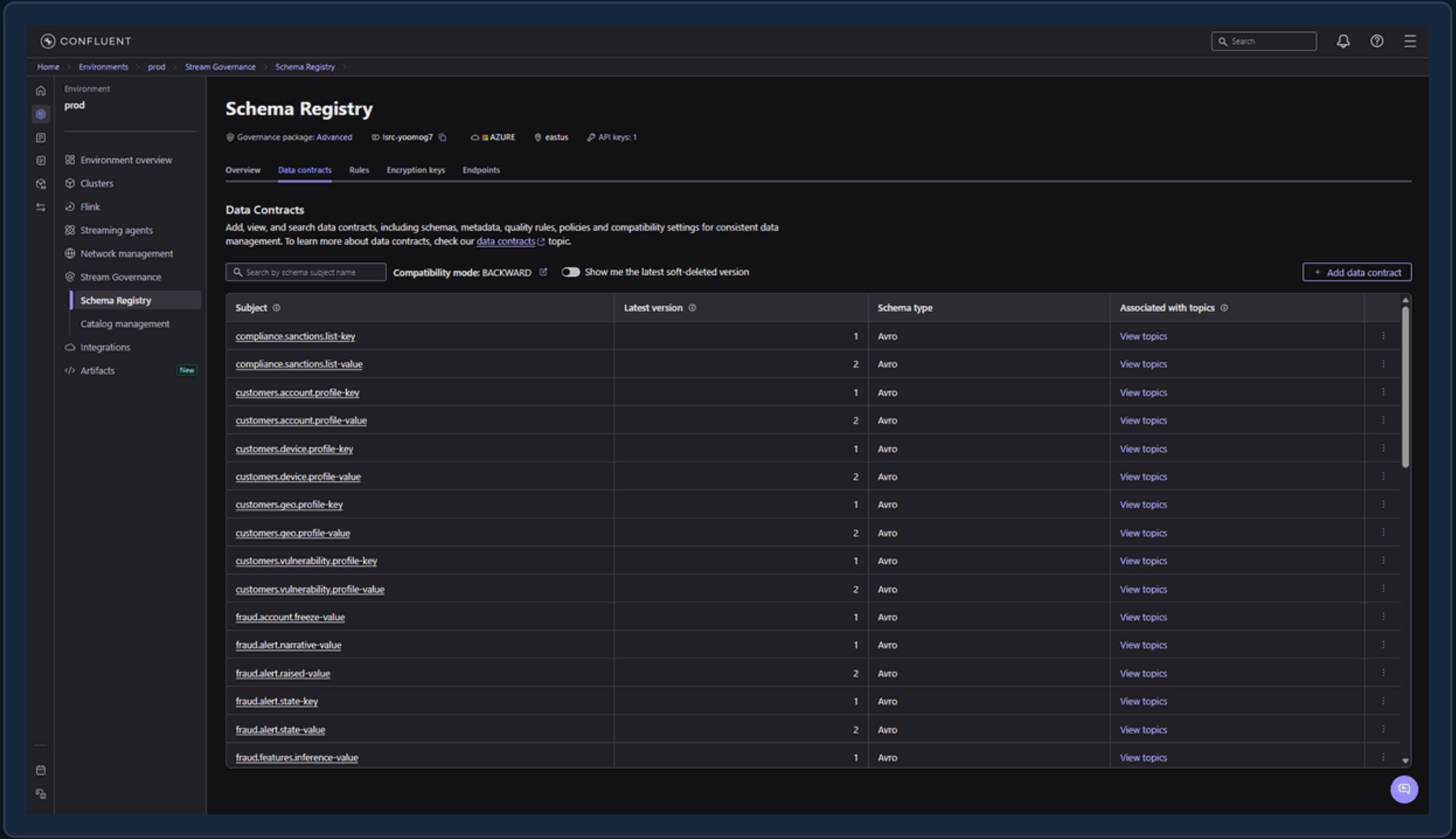
```
tableflow_topics = {
  "fraud.features.inference" = {
    enabled=true, table_formats=["ICEBERG"] }
}
```

Operational: fraud.alert. → Postgres (real-time) · Analytical: 4 topics → Iceberg (zero ETL)*

SCHEMA

Every topic is an Avro contract

LIVE



WHAT YOU'RE SEEING

- Subjects registered for both keys and values
- BACKWARD compatibility enforced on every evolution
- Lookup, alert and intelligence schemas all present
- Schema Registry links each contract to its stream

WHY IT MATTERS *No producer can publish a change that silently breaks a consumer — contracts are enforced, not hoped for.*

IBM STACK · FROM DEMO TO PRODUCTION

Enterprise Infrastructure.
Not a Demo Cluster.

⚡	**Confluent Cloud** The Hero — Kafka + Flink	Dedicated cluster · Private Link · Schema Registry · Stream Governance · 9 Flink SQL · Tableflow → Apache Iceberg
🏗	**HashiCorp Terraform** Infrastructure as Code	Two isolated roots · Zero UI resources · Topic naming enforced at plan time · Least-privilege RBAC · Dual-slot key rotation
☁️	**Microsoft Azure** Private Cloud Foundation	Private Link data plane · Key Vault (HMAC PII key) · Azure OpenAI GPT-4o · Postgres Flex Server · Container Apps
🤖	**IBM watsonx / Bob** AI-Assisted Engineering	6 design docs, 9 Flink statements, full IaC, Java producer — architected conversationally. AI operated the platform.
🏦	**RBIH MuleHunter.AI** Cross-Bank Mule Intelligence	Pure Kafka event contract — no REST, no shared DB · 90-day rolling state · consumer group mulehunter-rbih-cg · 40% weight

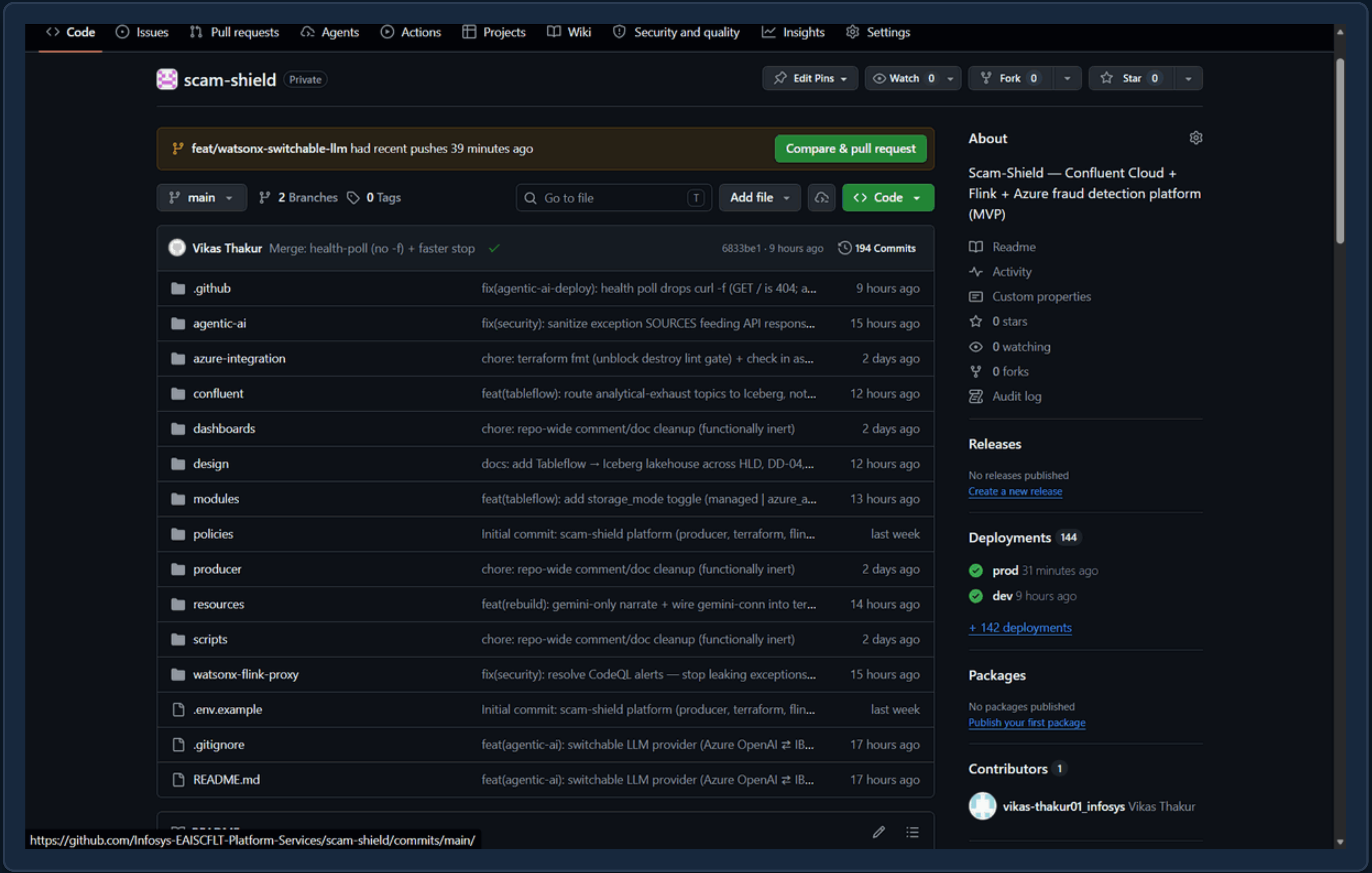
TERRAFORM · REPO

● LIVE

Everything is code, in one repo

WHAT YOU'RE SEEING

- One repo holds both roots: confluent/ and azure-integration/
- modules/, resources/ and policies/ sit beside app code
- No resource is ever created by hand in a console
- Drift is reconciled back to Git on every run

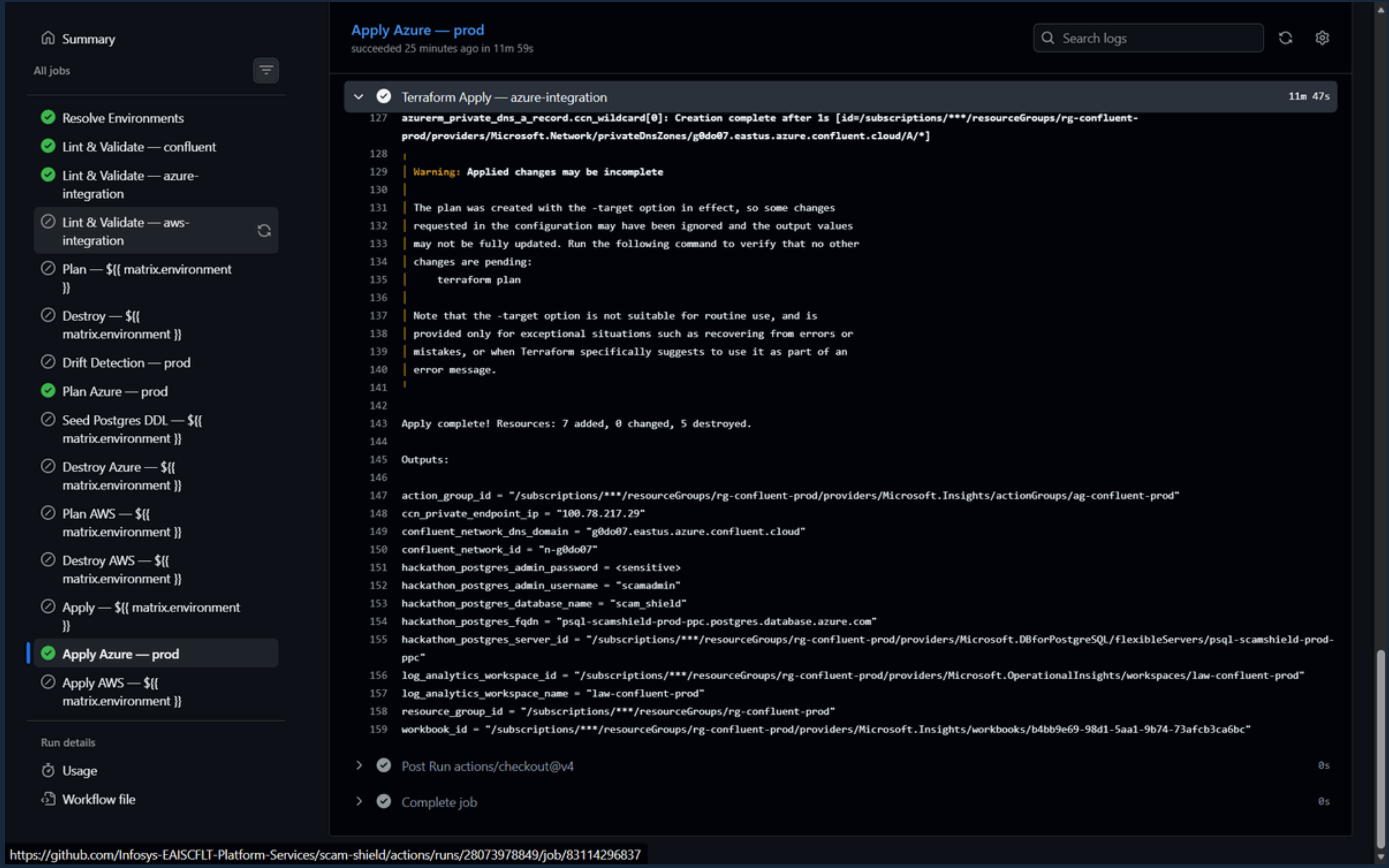


WHY IT MATTERS *Terraform is the sole source of truth — if it's not in Git, it doesn't exist.*

TERRAFORM · APPLY

One pipeline provisions Confluent and Azure

● LIVE



WHAT YOU'RE SEEING

- confluent/ applies first; azure-integration/ reads its state
- Key Vault, PrivateLink, Postgres and monitoring provisioned
- Apply consumes the approved plan with -auto-approve
- Outputs hand off cleanly between the two roots

WHY IT MATTERS *The whole platform — both clouds — stands up from code in one flow.*

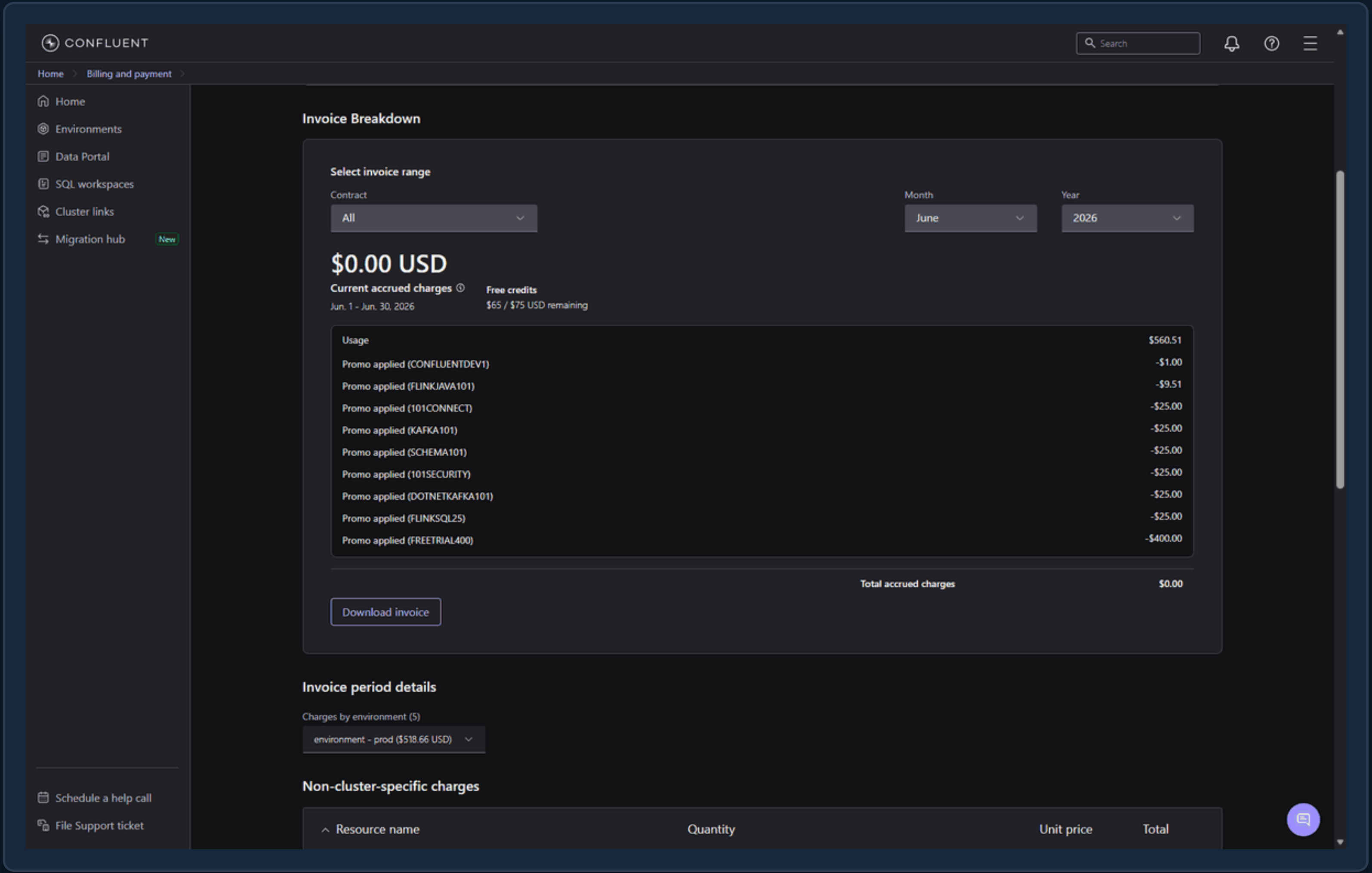
COST

LIVE

This enterprise stack runs on free credits

WHAT YOU'RE SEEING

- June 2026 accrued charges: $0.00
- ~$560 of usage fully offset by program promos
- Remaining free credits tracked in-console
- Transparent, itemised, per-resource billing



WHY IT MATTERS *Production-grade architecture demonstrated at effectively zero cost — and costed transparently for real planning.*

LIVE DEMO

Real Fraud. Real Decisions. Live on Confluent Cloud.

SIM Swap + APP Fraud

CHALLENGE

UPI P2P

STMT-RULES fires SIM_SWAP-001. STMT-CEPXR catches BENEFICIARY_ADD → AUTH → PAYMENT in 3s (XR dwell). HITL queued. OTP challenge auto-issued.

CNP + Geo-Impossible Travel

BLOCK

Debit Card

CNP fraud + geo-impossible (800km in 4 min) + IMEI mismatch. conf 0.88 ≥ 0.85 + amt ≤ ₹5L → auto-freeze. Card specialist AI dossier.

Mule Fan-In Network

HOLD

IMPS

MuleHunter CRITICAL (mule_prob 0.91). XR_MULE_CHAIN: 3 rapid CREDITs → DEBIT to new beneficiary in 8 min. Hard override → HOLD. Always HITL.

Structuring / AML

HARD BLOCK

NEFT

₹12L · 3 round-amount txns in 4h → CTR breached. Sanctions flag. sar_required=true. SAR auto-generated. ALL actions → HITL.

★ STMT-NARRATE fires post-decision for all 4 scenarios → AI Analyst Note (Gemini primary) appears on Fraud Ops Dashboard within seconds · Flink decides, AI explains, analyst reviews

DASHBOARD · COMMAND CENTER

A live command center for fraud ops

● LIVE

WHAT YOU'RE SEEING

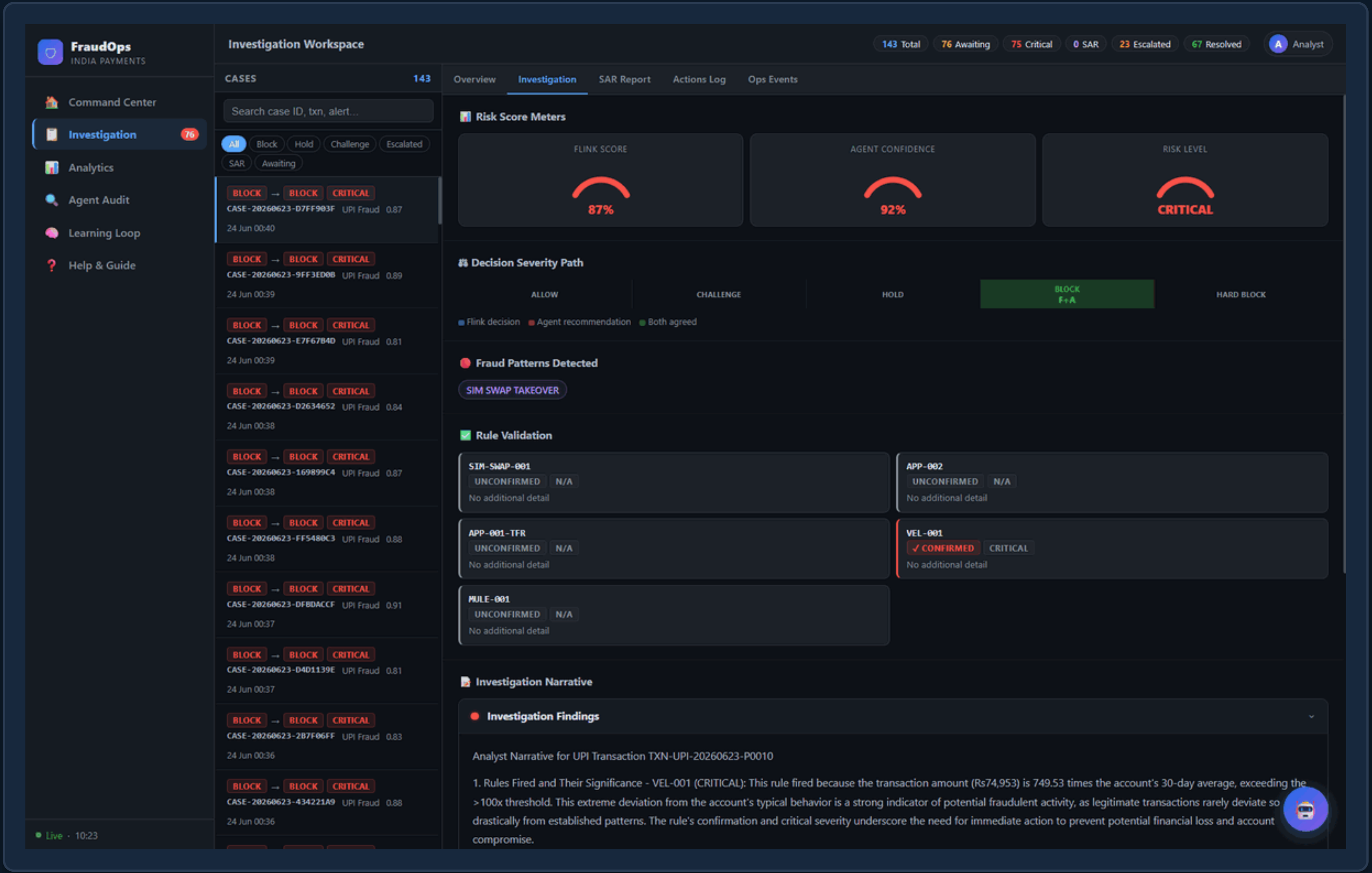
- Real-time tiles: total, awaiting analyst, critical, escalated
- 79% of cases auto-handled; the rest queued for a human
- Analyst queue ranked by risk, action and SLA
- Fed by fraud.alert.raised + platform.ops.event over SSE

WHY IT MATTERS *This is what the analyst sees — the whole fraud picture, with the few that matter surfaced first.*

DASHBOARD · INVESTIGATION

● LIVE

Deterministic score, agent cross-check



WHAT YOU'RE SEEING

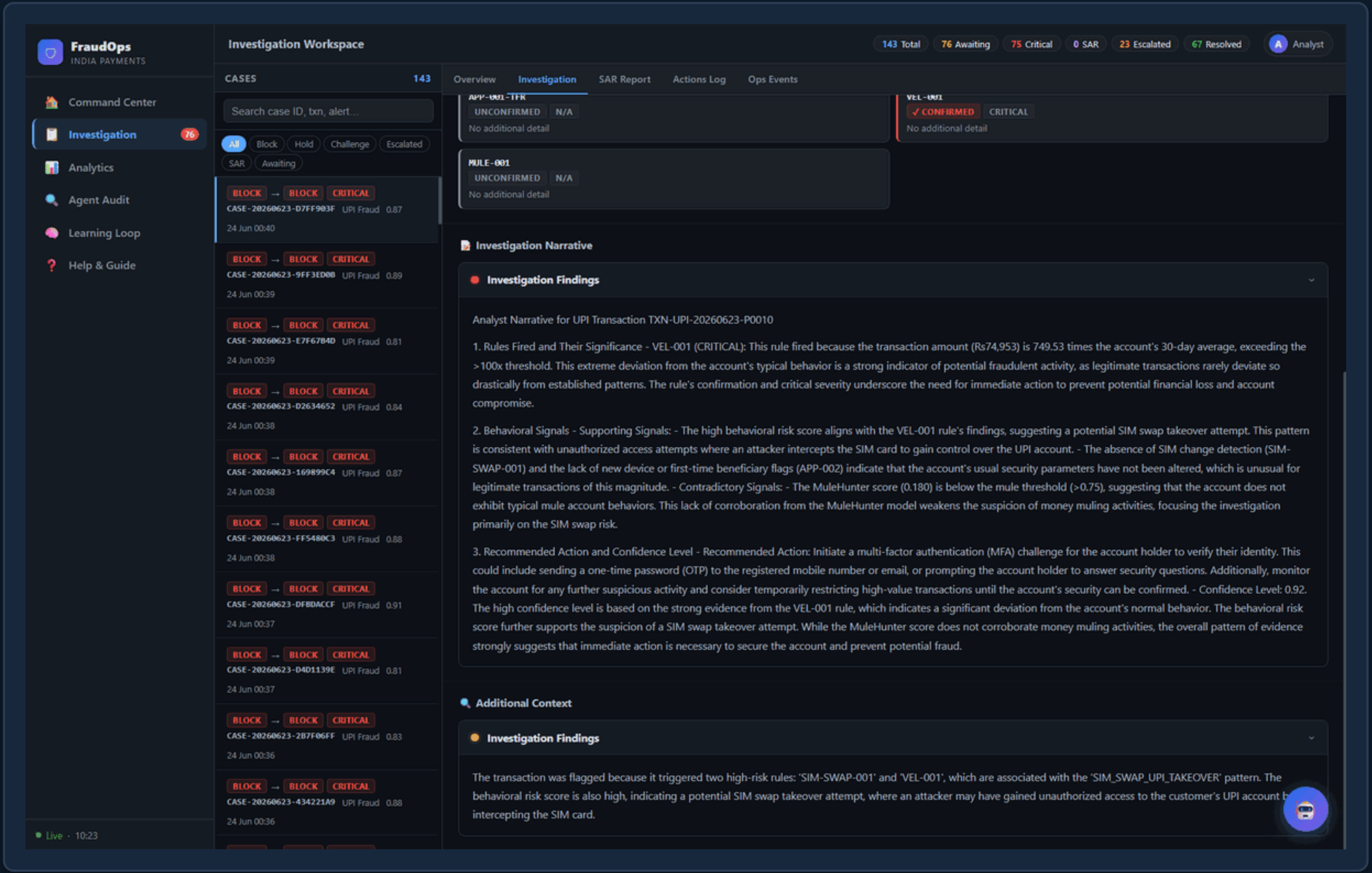
- Flink rule score and agent confidence side by side
- Decision severity path from ALLOW through HARD_BLOCK
- Detected fraud patterns and per-rule validation listed
- Python tools cross-validate Flink hits — never re-score

WHY IT MATTERS *The agent corroborates the deterministic decision; it never replaces it (INV-3).*

DASHBOARD · EXPLAINABILITY

● LIVE

Every alert gets a plain-English narrative



WHAT YOU'RE SEEING

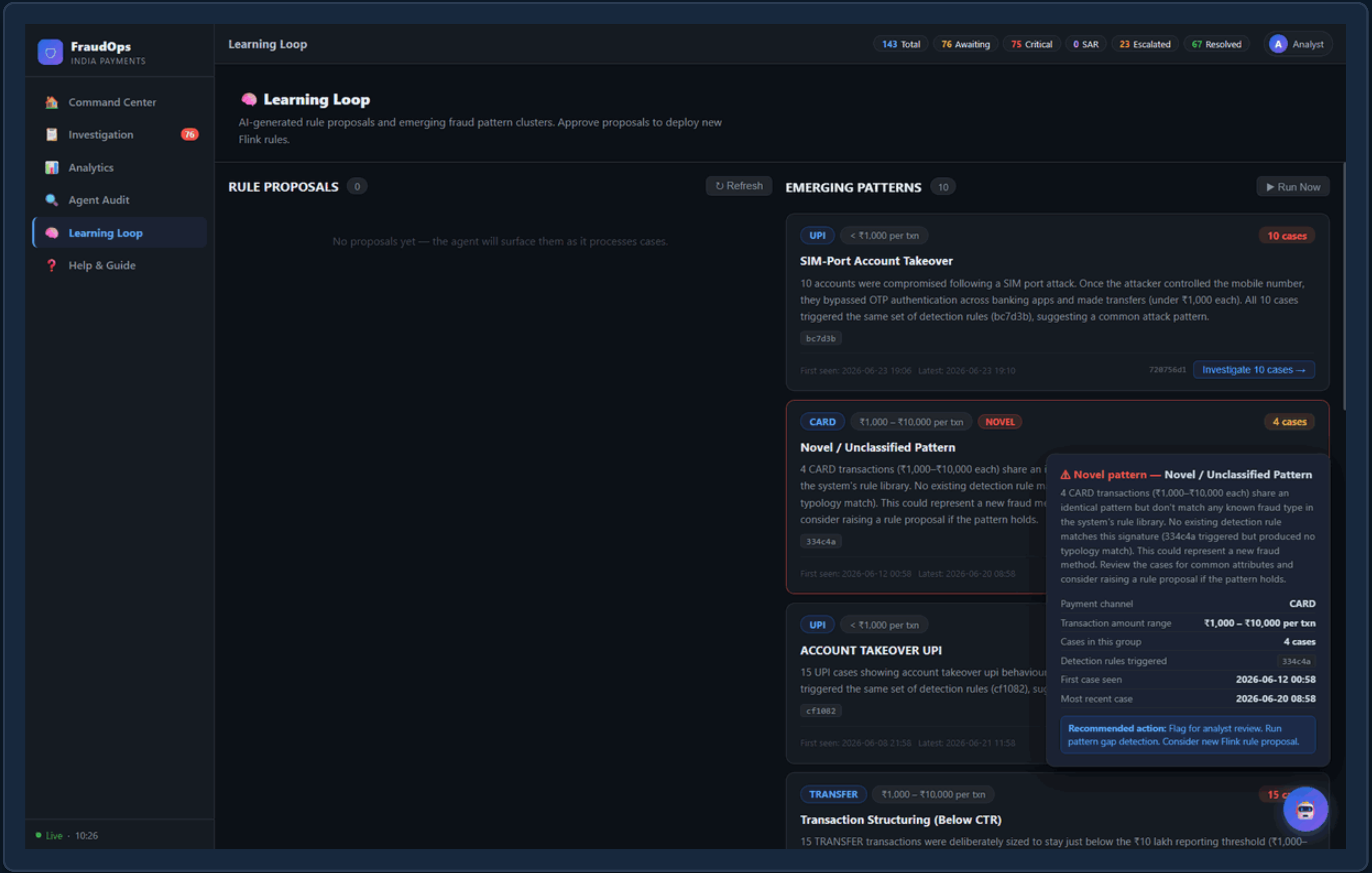
- A structured investigation narrative is drafted per case
- It cites the specific rules, signals and recommended action
- All model calls routed through the AI Governance gateway
- PII-free — the AI never sees raw identifiers (INV-1)

WHY IT MATTERS *Analysts read why, not just a score — exactly the analyst note promised in the pitch.*

DASHBOARD · LEARNING LOOP

• LIVE

The platform proposes its own new rules



WHAT YOU'RE SEEING

- Agents cluster resolved cases into emerging fraud patterns
- Each gap becomes a PENDING Flink SQL rule proposal
- An analyst must approve before anything deploys
- Approved rules ship as code via a pull request

WHY IT MATTERS *The system learns and suggests — humans still approve every rule change (INV-3).*

HACKATHON SUBMISSION

Not deploying a model.

Deploying the governance architecture that makes it safe.

PII-Safe by Architecture

HMAC-SHA256 at the producer. AI models never see raw PII — not by policy, by infrastructure. INV-1.

AI Never Decides

STMT-DECIDE: Flink rules + MuleHunter + AML. Deterministic, replayable, explainable. INV-3.

3-Backend Governed AI Overlay

STMT-NARRATE: Gemini (primary) / GPT-4o / WatsonX Granite via ML_PREDICT. All audited. INV-2.

Full Terraform IaC + Tableflow

Zero UI resources. Schema Registry. Tableflow → Iceberg. Demo-to-production in one codebase.

₹36,014 Cr Market. RBI-Ready.

PMLA / DPDP Act / INV-1-4. NCRP + DPIP integrations. 4 audit streams. 10-yr AML retention.

Connectors: PostgresCdcSourceV2 · PostgresSink (alerts) · PostgresSink (ops) | Flink: STMT-DECIDE + STMT-NARRATE | Schema: fraud.score.computed (Avro) | Lineage: raw→validated→enriched→scored→alert→narrative